A PROPER FAMILY OVER A DISCRETE VALUATION RING WITH REDUCED SPECIAL FIBRE IS REDUCED

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 7304-7311 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 7313-7339, the complete author proof (25 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Proper morphisms over a discrete valuation ring, reducedness of schemes, and the behaviour of components under the dominance hypothesis; the argument is retained.

Source reference. Original lemma 00IP: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7339–7339. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 05DB: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7339–7339. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to \mathrm{Spec}(R)$ be a proper morphism where R is a discrete valuation ring. Assume every irreducible component of X dominates $\mathrm{Spec}(R)$ (for example if f is flat) and assume the special fibre is reduced. Then both X and the generic fibre X_η are reduced.*

Proof. (The fact that a flat morphism satisfies the assumption on irreducible components follows for example from Divisors, Lemma 05DB or from going down for flat ring maps.) Assume the special fibre X_s is reduced. Let $x \in X$ be any point, and let us show that $\mathcal{O}_{X,x}$ is reduced; this will prove that X and X_η are reduced. Let $x \rightsquigarrow x'$ be a specialization with x' in the special fibre; such a specialization exists as a proper morphism is closed. Consider the local ring $A = \mathcal{O}_{X,x'}$. Then $\mathcal{O}_{X,x}$ is a localization of A, so it suffices to show that A is reduced. Let $\pi \in R$ be a uniformizer. If $a \in A$ is nonzero then there exists an $n \geq 0$ and an element $a' \in A$ such that $a = \pi^n a'$ and $a' \notin \pi A$. This follows from Krull intersection theorem (Algebra, Lemma 00IP). If a is nilpotent the a is contained in all minimal primes of A, and hence so is a', because π is not contained in any of the minimal primes of A by our assumption on the irreducible components of X. Thus a' is nilpotent.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

But a' maps to a nonzero element of the reduced ring $A/\pi A = \mathcal{O}_{X_s,x'}$. This is a contradiction unless A is reduced, which is what we wanted to show. $\square$